

ROLL NO:250701573

EXPERIMENT NO: 20 EXPERIMENT TO UNDERSATND PANDAS LIBRARY AND ITS
USE CASES IN DATA SCIENCE

PROGRAM:

```
import numpy as np

import pandas as pd

# Create DataFrame

list1=[[1,'Arun',85],[2,'Priya',92]]

df=pd.DataFrame(list1)

df.columns=["ID","Name","Marks"]

df

df.info()

# --- ADDED THIS LINE TO GENERATE THE FILESOITDOES NOT FAIL ---
df.to_csv("Students.csv", index=False)

# Read CSV (Replaces the above df)

df=pd.read_csv("Students.csv")

df

df.info()

df.head()

df.tail()

# Column Operations

df["Marks"]

type(df["Marks"])

df["Marks"].mean()

df["Marks"].median()

df["Marks"].mode()

df["Marks"].var()
```

```

df["Marks"].std()

# Descriptions
df.describe()

df.describe(include='all')

# Array Conversion & Rebuilding
empcol=df.columns

empcol

emparray=df.values

emparray

Student_df=pd.DataFrame(emparray,columns=empcol)

Student_df

```

OUTPUT:

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0    ID      2 non-null      int64
1   Name     2 non-null      object
2   Marks    2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0    ID      2 non-null      int64
1   Name     2 non-null      object
2   Marks    2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes

```

	ID	Name	Marks
0	1	Arun	85
1	2	Priya	92